

MAT1252

Mathematics for Computing

Lecture 03

Boolean Algebra & Logic Circuits

Outline

01. Boolean Algebra
02. Evaluating Boolean expressions
03. Logic gates
04. From logic circuit to Boolean expression
05. From Boolean expression to logic circuit
06. Truth tables and Boolean expressions
07. From sums of products to truth tables
08. Fundamental products
09. Laws of Boolean algebra
10. Using the laws
11. Some examples using de Morgan's laws
12. Expanding a SOP into a complete sum of products



Lecture's Major Objectives

After completing this section, students should be able to:

- use truth tables for operators AND, OR and NOT
- evaluate Boolean expressions involving these operators
- construct a truth table for a given sum of products
- given a logic circuit, write down a corresponding Boolean expression
- given a Boolean expression, draw a corresponding logic circuit
- use the laws of Boolean algebra to simplify an expression to a sum of products
- expand a sum of products as a complete sum of products
- given a truth table, write down the corresponding complete sum of products

Boolean Algebra

In Boolean Algebra there are just two values, 0 and 1, and three operations, "multiplication" ($\bullet$), "addition" (+) and "complementation" (').

The laws for these operations are as shown below. Some of the laws look like ordinary arithmetic.

Laws for multiplication:

$$0 \bullet 0 = 0 \quad 0 \bullet 1 = 0 \quad 1 \bullet 0 = 0 \quad 1 \bullet 1 = 1$$

Laws for addition:

$$0 + 0 = 0 \quad 0 + 1 = 1 \quad 1 + 0 = 1 \quad 1 + 1 = 1$$

Laws for complementation:

$$0' = 1 \quad 1' = 0$$

Boolean Algebra

The laws for Boolean Algebra look identical to the truth tables for the logical operators OR $\vee$, AND $\wedge$ and NOT $\sim$:

p	q	$p + q$
0	0	0
0	1	1
1	0	1
1	1	1

p	q	$p \cdot q$
0	0	0
0	1	0
1	0	0
1	1	1

p	p'
0	1
1	0

The correspondence is:

$0 \leftrightarrow F$ $1 \leftrightarrow T$ $+ \leftrightarrow \vee$ $\cdot \leftrightarrow \wedge$ $' \leftrightarrow \sim$

Boolean Algebra

One major advantage of using a notation with symbols " $\cdot$ " and " $+$ " is that many of the laws of logic look like the familiar laws of ordinary algebra, and that makes algebraic manipulation so much easier.

Order of operations:

- "complement ($'$)" takes precedence over "multiplication ($\cdot$)"
- "multiplication ($\cdot$)" takes precedence over "addition ($+$)"

unless indicated otherwise by means of brackets.

Boolean Algebra

Example: Evaluate the Boolean expression $x + y' \cdot z$, in the case where $x = 1$, $y = 0$, $z = 0$.

Solution: Based on order of operations, first find y' and then $y' \cdot z$, and finally $x + y' \cdot z$.

In other words we **first perform** the "complement (')", then the "multiplication ($\cdot$)" and **finally** the "addition (+)".

$$\text{So } x + y' \cdot z = 1 + ((0') \cdot 0) = 1 + (1 \cdot 0) = 1 + 0 = \mathbf{1}.$$

Boolean Algebra

We often drop the symbol "•" altogether. So, for example,

the distributive law

$$x \bullet (y + z) = (x \bullet y) + (x \bullet z)$$

may be written as

$$x(y + z) = xy + xz$$

NOTE: When carrying out some algebraic operations, such as using de Morgan's laws, or when working with numerical values, it is often a good idea to **temporarily** use the symbol "•".

Example using de Morgan's laws:

$$(xy'z)' = (x \bullet y' \bullet z)' = x' + (y')' + z' = x' + y + z'$$

Evaluating Boolean expressions

Example 1:

If $x = 1$, $y = 0$, $z = 0$, find the value of $xy' + yz'$.

Evaluating Boolean expressions

Example 1:

If $x = 1$, $y = 0$, $z = 0$, find the value of $xy' + yz'$.

Solution

$$xy' + yz' = (1 \cdot 0') + (0 \cdot 0') = (1 \cdot 1) + (0 \cdot 1) = 1 + 0 = 1$$

Evaluating Boolean expressions

Example 2:

If $x = 0$ and $y = 1$, evaluate $(x' + y)'$.

Evaluating Boolean expressions

Example 2:

If $x = 0$ and $y = 1$, evaluate $(x' + y)'$.

Solution

$$(x' + y)' = (0' + 1)' = (1 + 1)' = 1' = 0$$

Evaluating Boolean expressions

Example 3:

If $x = 0$, $y = 1$ and $z = 0$ find the value of $(x + yz')(xy' + z)$.

Evaluating Boolean expressions

Example 3:

If $x = 0$, $y = 1$ and $z = 0$ find the value of $(x + yz')(xy' + z)$.

Solution

$$\begin{aligned}(x + yz')(xy' + z) &= (0 + 1 \cdot 0')(0 \cdot 1' + 0) \\&= (0 + 1 \cdot 1)(0 \cdot 0 + 0) \\&= (0 + 1)(0 + 0) \\&= 1 \cdot 0 \\&= 0\end{aligned}$$

Evaluating Boolean expressions

Example 4:

If $x = 0$, $y = 1$, $z = 0$ find the value of $xyz + x'yz' + x'y'z$.

Evaluating Boolean expressions

Example 4:

If $x = 0$, $y = 1$, $z = 0$ find the value of $xyz + x'yz' + x'y'z$.

Solution

$$\begin{aligned}xyz + x'yz' + x'y'z &= 0 \cdot 1 \cdot 0 + 0' \cdot 1 \cdot 0' + 0' \cdot 1' \cdot 0 \\&= 0 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 0 \cdot 0 \\&= 0 + 1 + 0 \\&= 1\end{aligned}$$

Sums of products

Two very important concepts that we use in Boolean algebra are **products** and **sums of products**.

- A **product** is an expression in which the variables are **multiplied**

e.g. $xy'z$, $x'y$ or yz'

- A **sum of products** is an expression in which **products** are added

e.g. $xy + y'z + x'yz'$ or $x'y' + xy'z + xz$

Evaluating Boolean expressions

Exercise

For each expression, what values of **x**, **y** and **z** will make the product equal to **1**?

xyz' **x** = **y** = **z** =

x'y'z **x** = **y** = **z** =

x'yz **x** = **y** = **z** =

xy'z' **x** = **y** = **z** =

Evaluating Boolean expressions

Exercise

For each expression, what values of **x**, **y** and **z** will make the expression equal to **1**?

xyz'	$x = 1$	$y = 1$	$z = 0$
$x'y'z$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$
$x'yz$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$
$xy'z'$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$

Evaluating Boolean expressions

Exercise

For each expression, what values of **x**, **y** and **z** will make the expression equal to **1**?

xyz'	$x = 1$	$y = 1$	$z = 0$
$x'y'z$	$x = 0$	$y = 0$	$z = 1$
$x'yz$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$
$xy'z'$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$

Evaluating Boolean expressions

Exercise

For each expression, what values of **x**, **y** and **z** will make the expression equal to **1**?

xyz'	$x = 1$	$y = 1$	$z = 0$
$x'y'z$	$x = 0$	$y = 0$	$z = 1$
$x'yz$	$x = 0$	$y = 1$	$z = 1$
$xy'z'$	$x = \dots\dots\dots$	$y = \dots\dots\dots$	$z = \dots\dots\dots$

Evaluating Boolean expressions

Exercise

For each expression, what values of **x**, **y** and **z** will make the expression equal to **1**?

xyz'	$x = 1$	$y = 1$	$z = 0$
--------	---------	---------	---------

$x'y'z$	$x = 0$	$y = 0$	$z = 1$
---------	---------	---------	---------

$x'yz$	$x = 0$	$y = 1$	$z = 1$
--------	---------	---------	---------

$xy'z'$	$x = 1$	$y = 0$	$z = 0$
---------	---------	---------	---------

Evaluating Boolean expressions

Exercise

For each of the following combinations of values, which product of **x**, **y** and **z** (with or without complements) is equal to **1**?

x = 0, y = 1, z = 0

.....

x = 1, y = 0, z = 1

.....

x = 1, y = 1, z = 1

.....

x = 0, y = 0, z = 0

.....

Evaluating Boolean expressions

Exercise

For each of the following combinations of values, which product of **x**, **y** and **z** (with or without complements) is equal to **1**?

x = 0, y = 1, z = 0

x'y z'

x = 1, y = 0, z = 1

.....

x = 1, y = 1, z = 1

.....

x = 0, y = 0, z = 0

.....

Evaluating Boolean expressions

Exercise

For each of the following combinations of values, which product of **x**, **y** and **z** (with or without complements) is equal to **1**?

x = 0, y = 1, z = 0

x'yz'

x = 1, y = 0, z = 1

xy'z

x = 1, y = 1, z = 1

.....

x = 0, y = 0, z = 0

.....

Evaluating Boolean expressions

Exercise

For each of the following combinations of values, which product of **x**, **y** and **z** (with or without complements) is equal to **1**?

x = 0, y = 1, z = 0

x'yz'

x = 1, y = 0, z = 1

xy'z

x = 1, y = 1, z = 1

xyz

x = 0, y = 0, z = 0

.....

Evaluating Boolean expressions

Exercise

For each of the following combinations of values, which product of **x**, **y** and **z** (with or without complements) is equal to **1**?

$$x = 0, y = 1, z = 0$$

$$x'yz'$$

$$x = 1, y = 0, z = 1$$

$$xy'z$$

$$x = 1, y = 1, z = 1$$

$$xyz$$

$$x = 0, y = 0, z = 0$$

$$x'y'z'$$

Logic gates

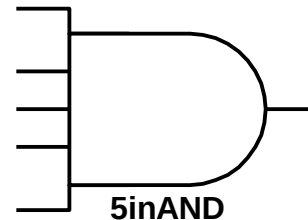
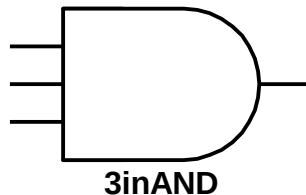
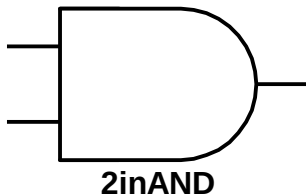
A logic gate is an **electronic device** which implements Boolean operations.

It has one or more inputs each of which will accept a "low" voltage (input = 0) or a "high" voltage (input = 1).

Logic gates

AND gate

An AND gate has two or more inputs, and its output is the **product** of the inputs.

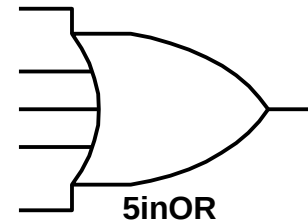
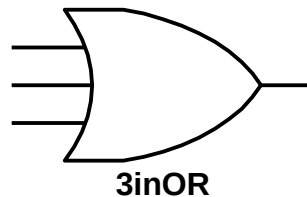
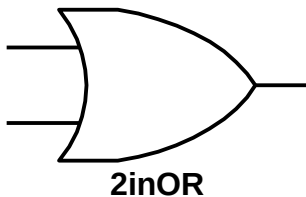


The output of an AND is 1 if all the inputs are 1; otherwise the output is 0.

Logic gates

OR gate

An OR gate has two or more inputs, and its output is the **Boolean sum** of the inputs.

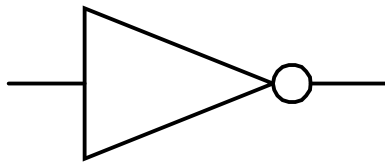


The output of an OR is 0 if all the inputs are 0; otherwise the output is 1.

Logic gates

NOT gate

A NOT gate (also called an inverter) has one input and one output.



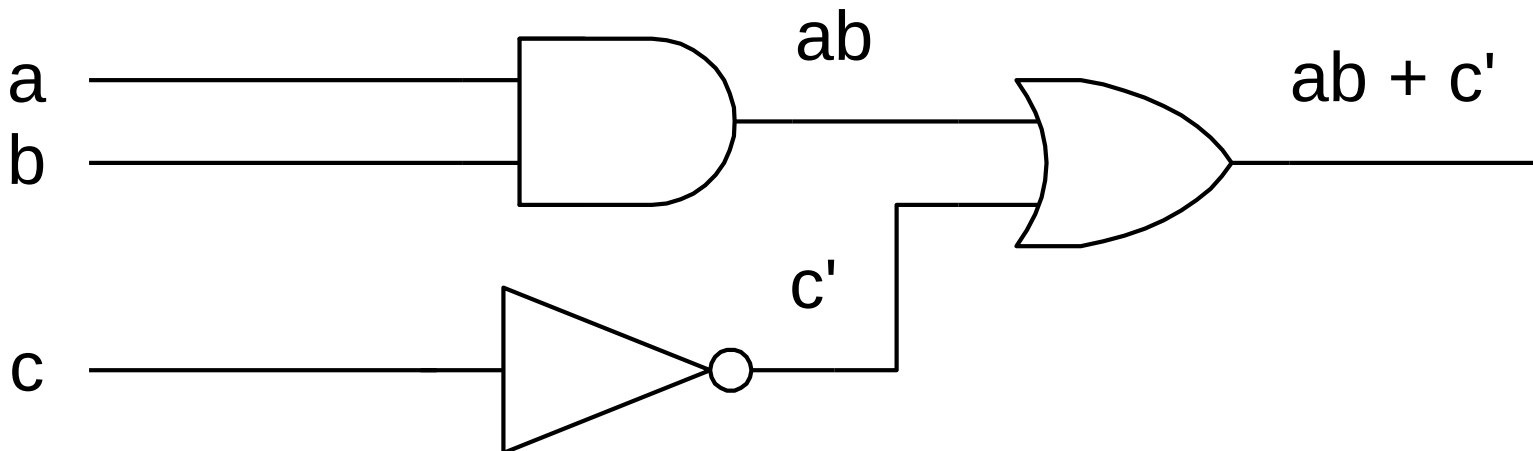
The output of a NOT is the complement (opposite) of the input.

In other words, an input of 1 gives output of 0 and vice versa.

From logic circuit to Boolean expression

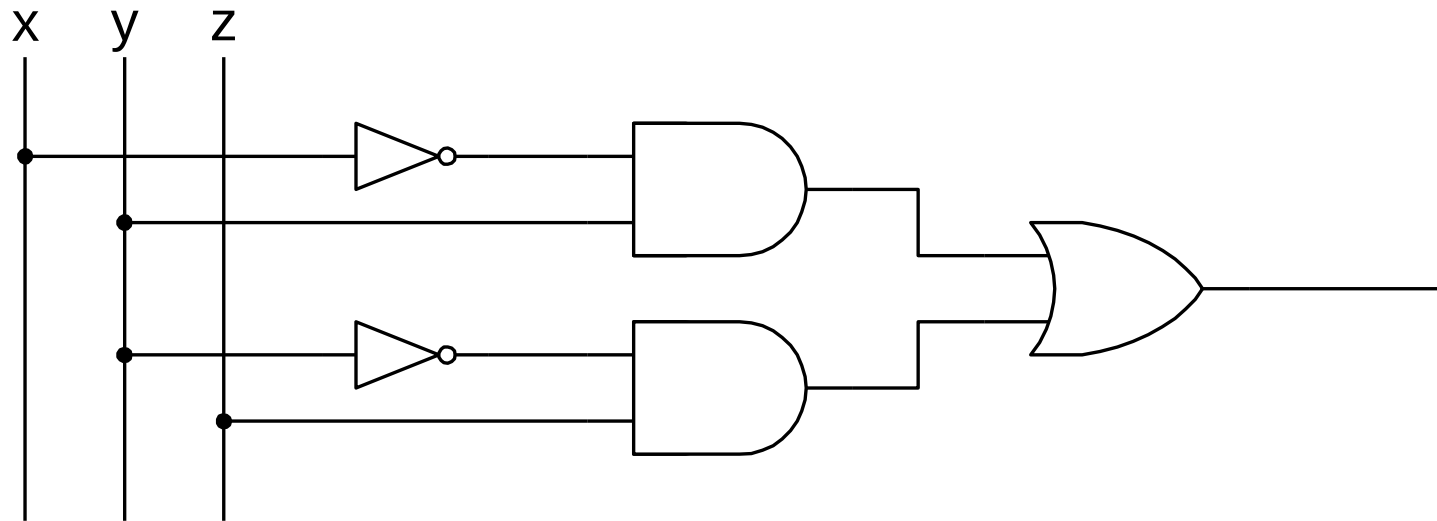
The Boolean expression for a particular circuit can be determined by following the output of each logic gate as it connects to the input of another logic gate.

It can be helpful to write the appropriate expression at the output of each gate, as shown in the following example:



From logic circuit to Boolean expression

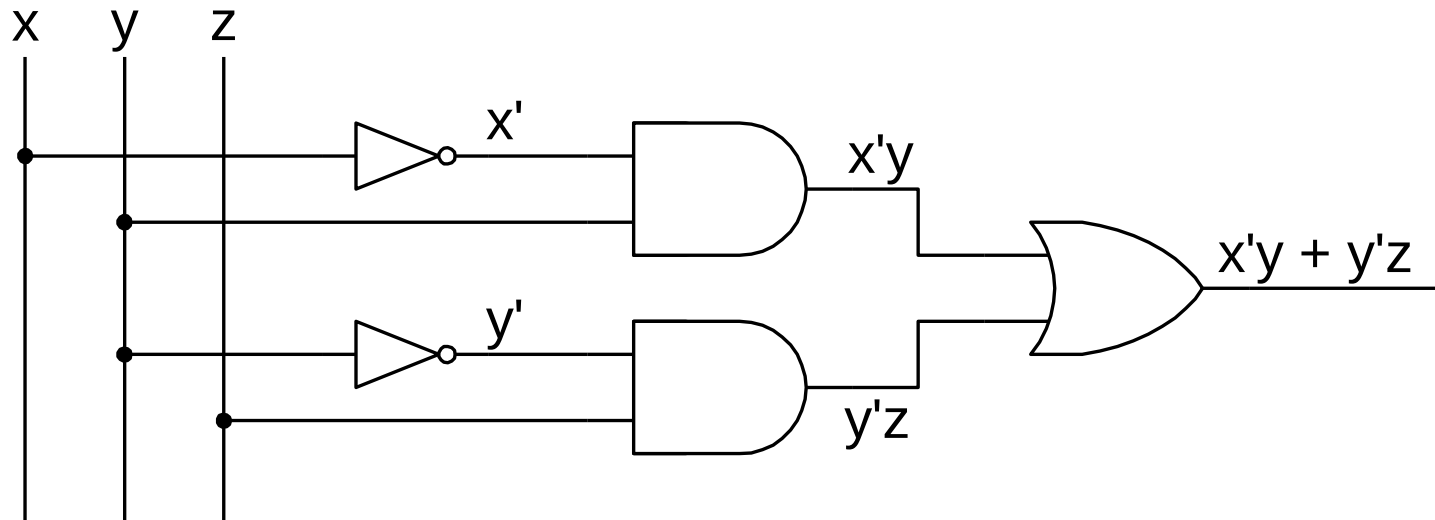
Example 1



From logic circuit to Boolean expression

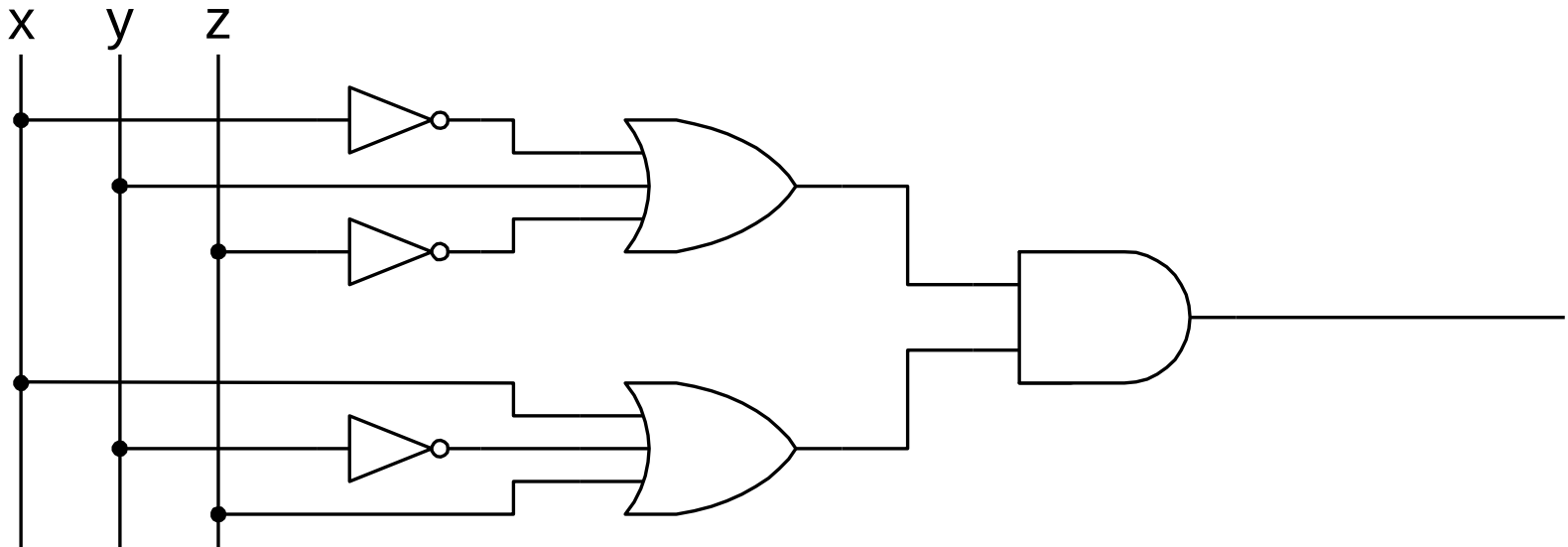
Example 1

Solution



From logic circuit to Boolean expression

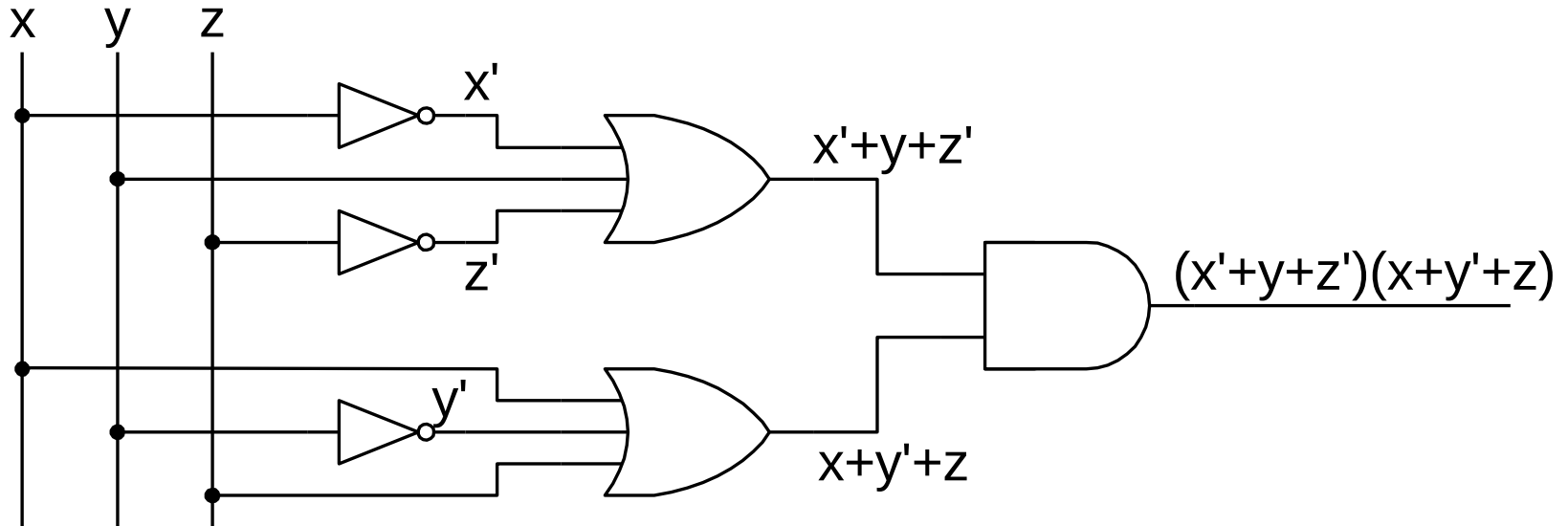
Example 2



From logic circuit to Boolean expression

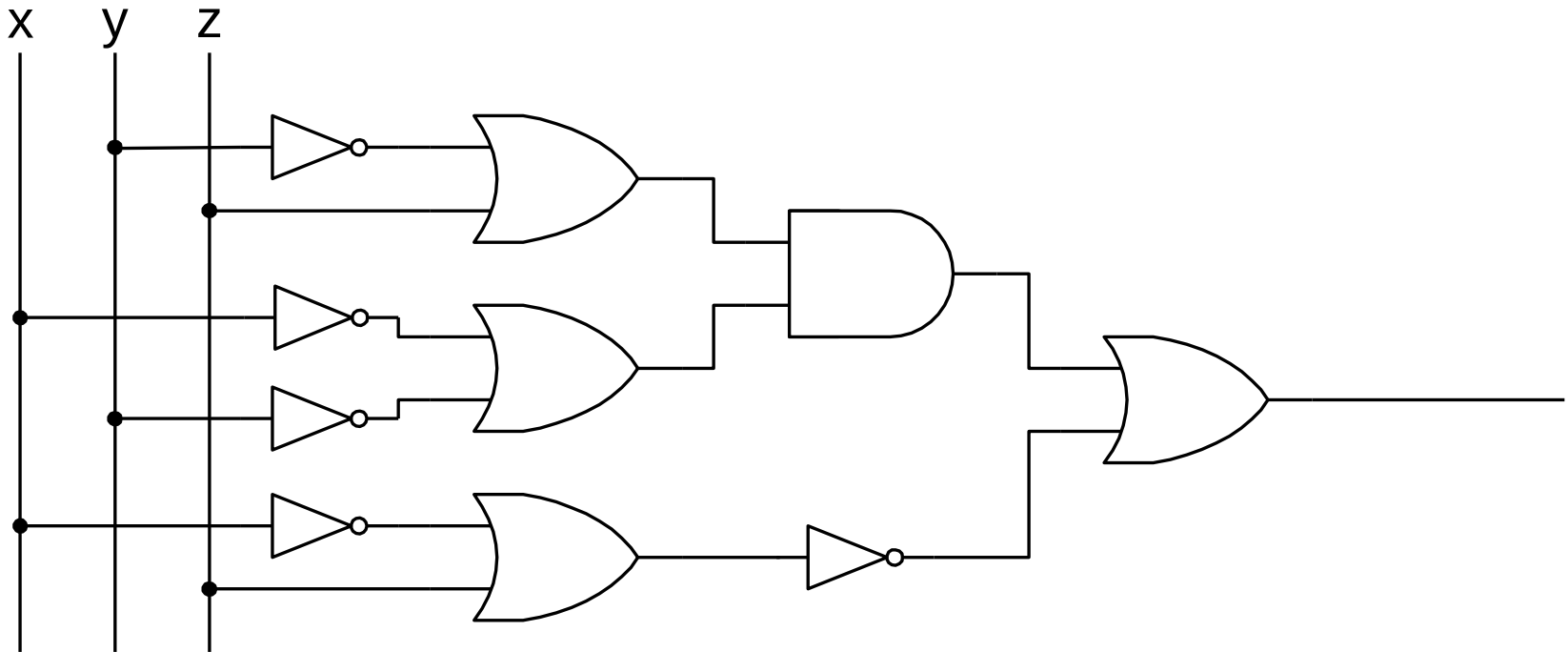
Example 2

Solution



From logic circuit to Boolean expression

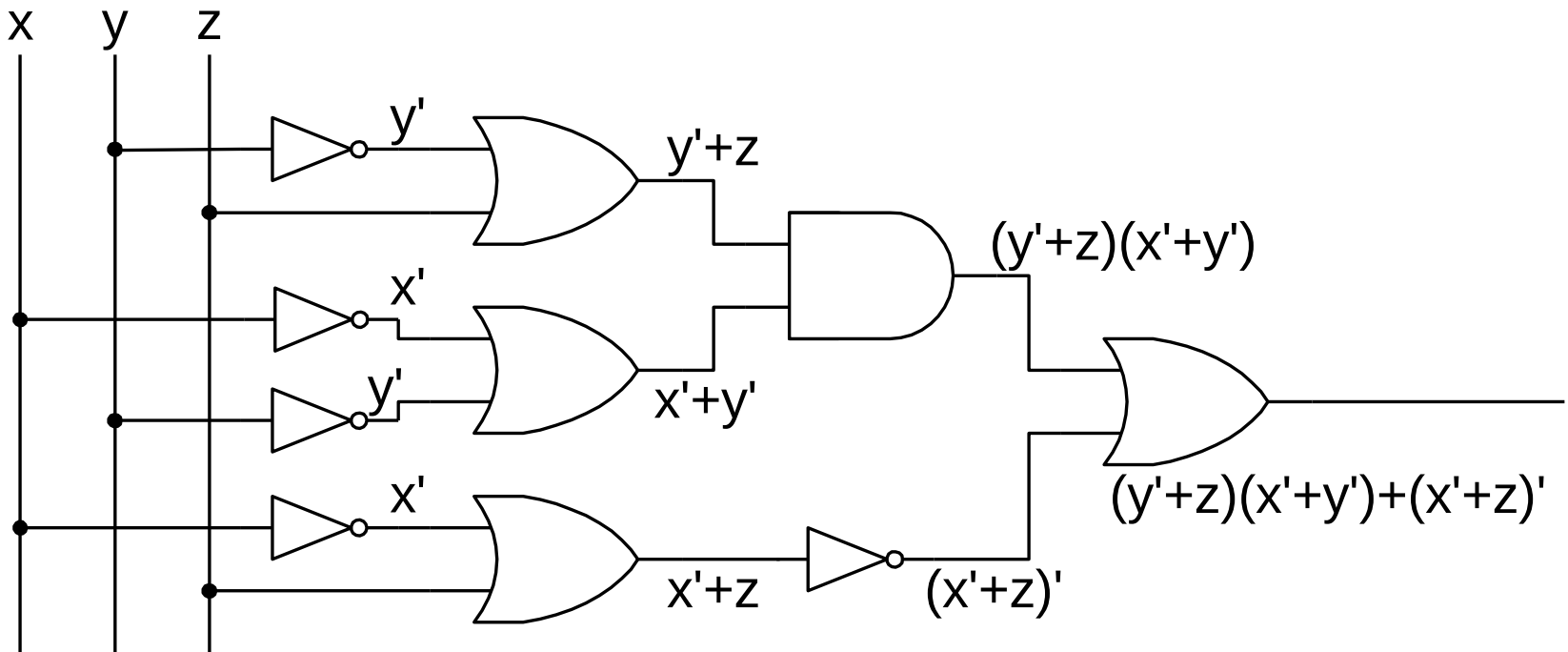
Example 3



From logic circuit to Boolean expression

Example 3

Solution



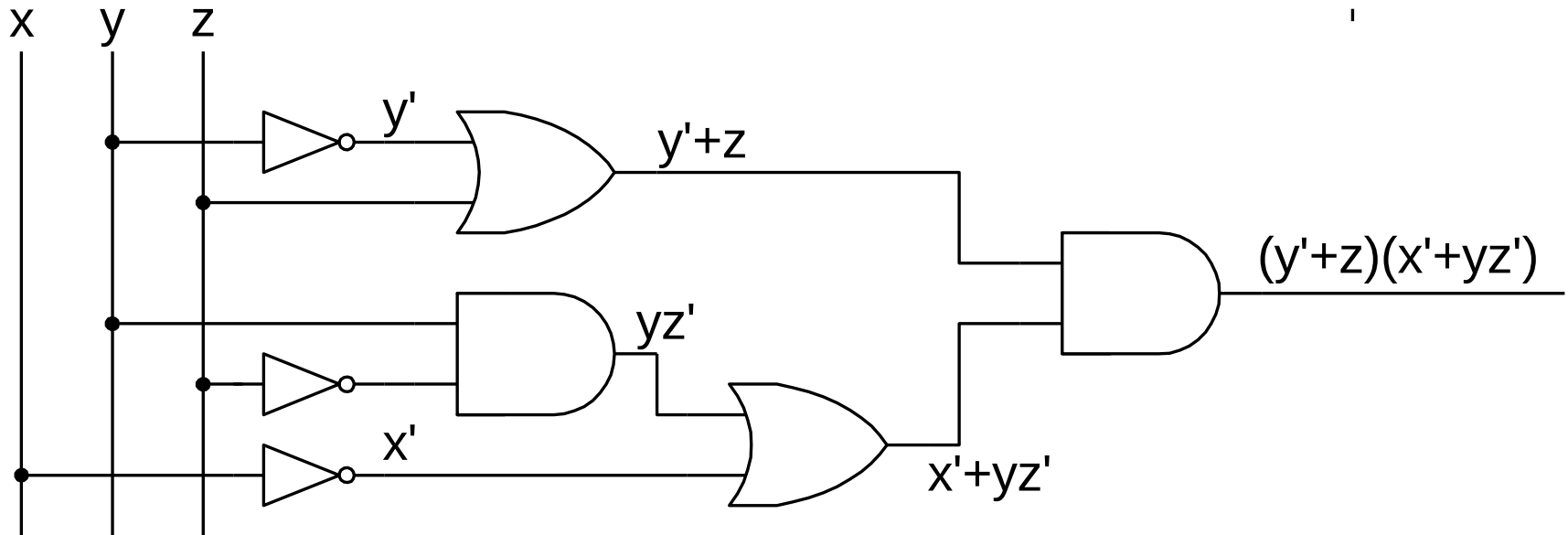
From Boolean expression to logic circuit

Example 1 $(y' + z)(x' + yz')$

From Boolean expression to logic circuit

Example 1 $(y' + z)(x' + yz')$

Solution



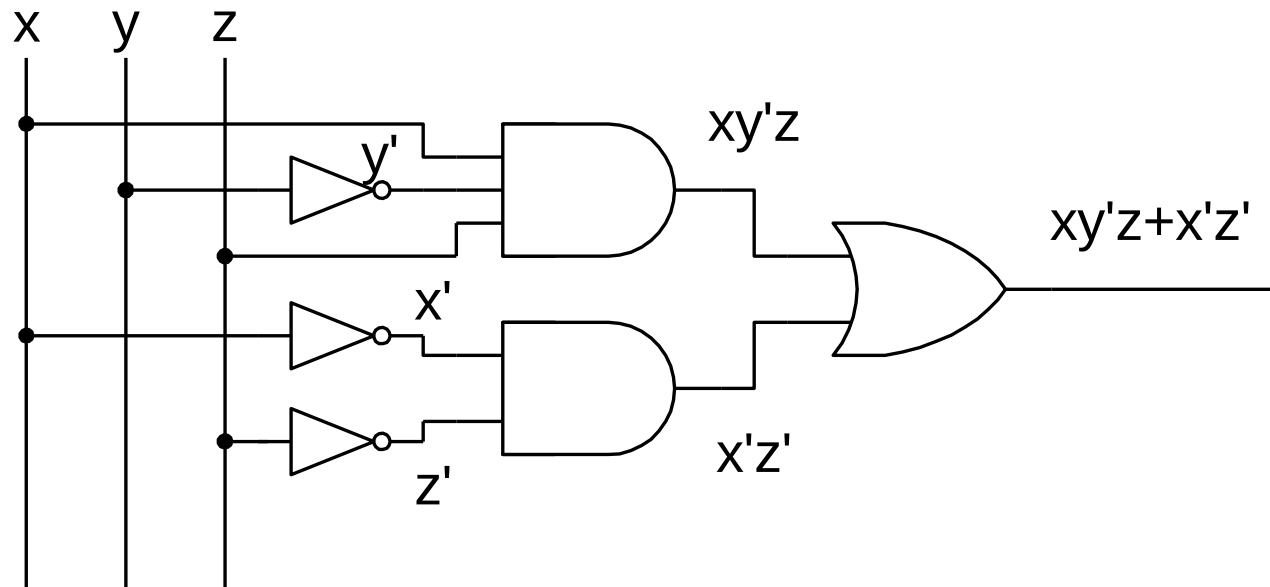
From Boolean expression to logic circuit

Example 2 $xy'z + x'z'$

From Boolean expression to logic circuit

Example 2 $xy'z + x'z'$

Solution



From Boolean expression to logic circuit

Example 3

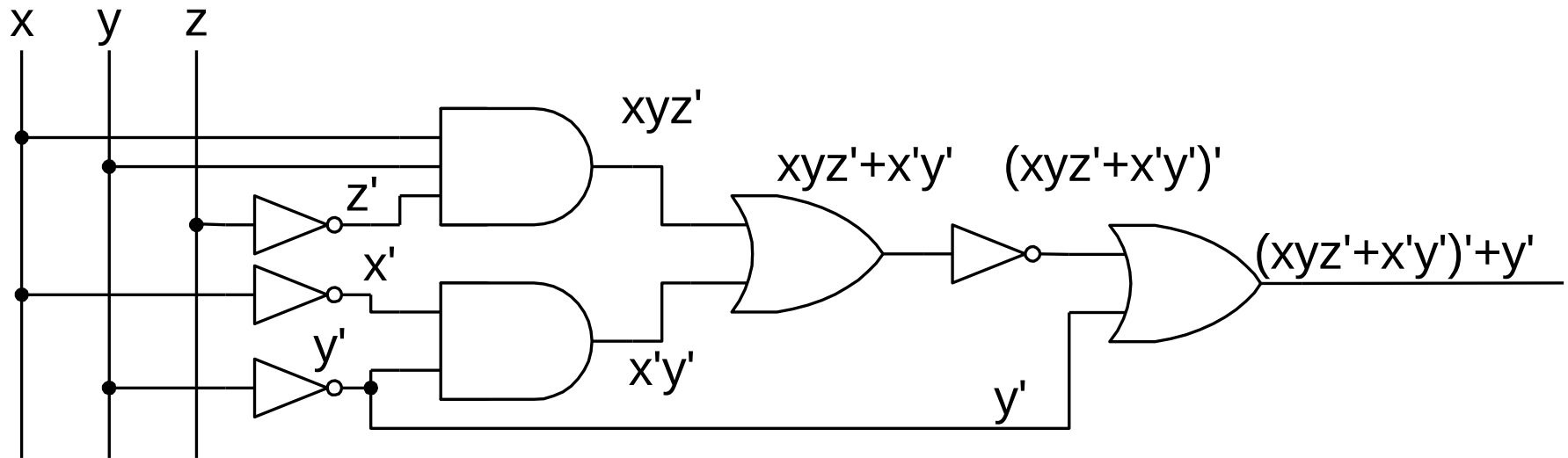
$$(xyz' + x'y')' + y'$$

From Boolean expression to logic circuit

Example 3

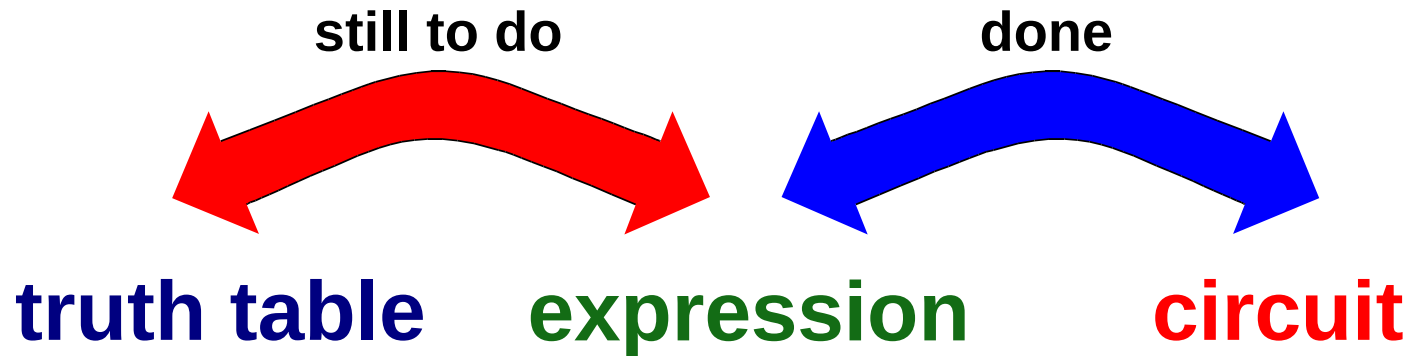
$$(xyz' + x'y')' + y'$$

Solution



Truth tables and Boolean expressions

It is important to be able to transfer between truth tables, Boolean expressions and circuits. The "links" are shown below.



We have already looked at the "links" between expressions and circuits.

Truth tables and Boolean expressions

There are two points to note:

- In constructing truth tables for expressions, we will confine our attention to the "sums of products" since these correspond to the AND-OR circuits.
- The link from truth tables to expressions is important for the design of circuits, and here we will need the concept of "fundamental product".

From sums of products to truth tables

Recall that **sums of products** are expressions such as

$$xy + y'z \quad \text{or} \quad x'y'z + yz' + xy'z'$$

- Products: combined using AND
- Sums of products: products combined using OR

To construct a truth table for a sum of products, we use **one column for each product** and then **one for the sum of products**, as in the examples.

From sums of products to truth tables

Example

Find the truth table for $L = x'yz + xy' + x'z'$

x	y	z	$x'yz$	xy'	$x'z'$	L
0	0	0	0	0	1	1
0	0	1	0	0	0	0
0	1	0	0	0	1	1
0	1	1	1	0	0	1
1	0	0	0	1	0	1
1	0	1	0	1	0	1
1	1	0	0	0	0	0
1	1	1	0	0	0	0

From sums of products to truth tables

Example

Find the truth table for $P = xyz' + x'y'$

x	y	z	xyz'	$x'y'$	P
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	0	0	0

Fundamental products

In order to find a Boolean expression corresponding to a given truth table, we need the concept of **fundamental product**. In a fundamental product, **each variable must be present**, with or without a complement.

For just **two** variables **x** and **y**, there are **four fundamental products**.

They are:

xy , xy' , $x'y$ and $x'y'$

For **three** variables **x**, **y** and **z**, there are **eight** fundamental products:

xyz , xyz' , $xy'z$, $xy'z'$, $x'yz$, $x'yz'$, $x'y'z$, and $x'y'z'$

Fundamental products

For four variables w , x , y and z , there are sixteen fundamental products:

$wxyz$, $wxyz'$, $wxy'z$, ..., $w'x'y'z'$

In general, if there are n variables, there will be 2^n fundamental products.

Note that if we are dealing with the three variables x , y and z , $x'yz$ is one of the fundamental products but not if we are dealing with the four variables w , x , y and z because the variable w is missing.

Fundamental products

Each fundamental product equals to 1 for one and only one combination of the variables.

Suppose we are dealing with three variables x , y and z .

$xy'z'$ is equal to 1 if and only if $x = 1$, $y = 0$ and $z = 0$.

$xy'z$ is equal to 1 if and only if $x = 1$, $y = 0$ and $z = 1$.

From truth table to Boolean expression

Example

Find a Boolean expression corresponding to the truth table:

x	y	z	L
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

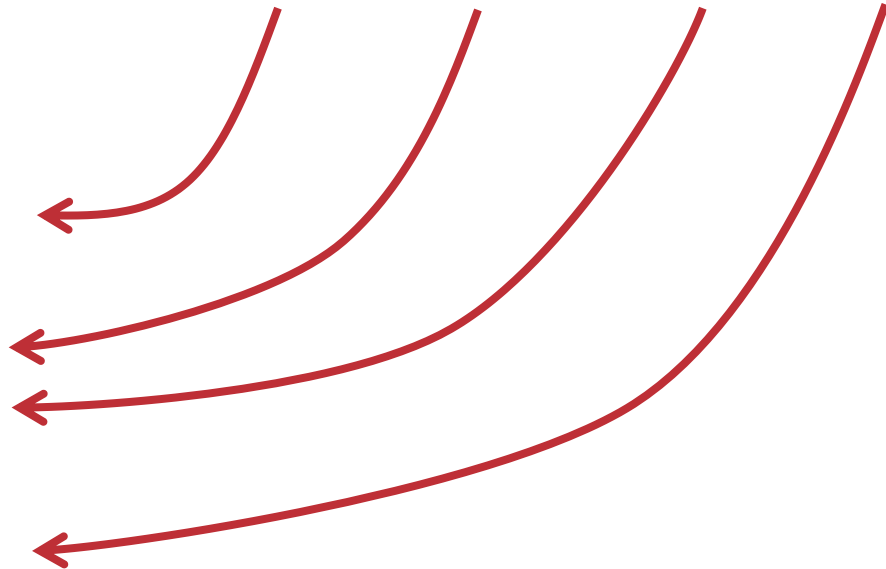
From truth table to Boolean expression

Example

Find a Boolean expression corresponding to the truth table:

x	y	z	L
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$$L = x'yz' + xy'z' + xy'z + xyz$$



From truth table to Boolean expression

Exercise (you try!)

Find a Boolean expression corresponding to the truth table:

a	b	c	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

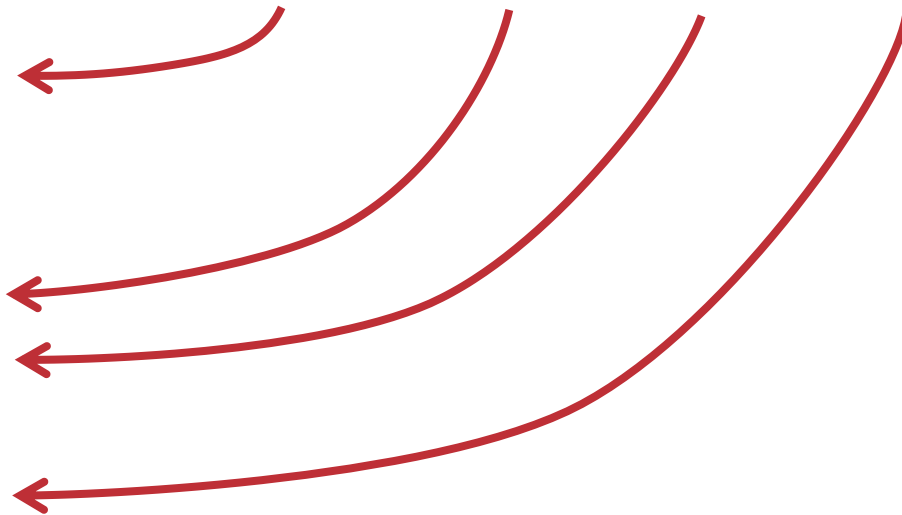
From truth table to Boolean expression

Exercise (you try!)

Find a Boolean expression corresponding to the truth table:

a	b	c	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$$P = a'b'c' + a'bc + ab'c' + abc'$$



From truth table to Boolean expression

w	x	y	z	L
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Exercise

Find a Boolean expression corresponding to this truth table.

From truth table to Boolean expression

w	x	y	z	L
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Exercise

Find a Boolean expression corresponding to this truth table.

$$L = w'x'yz' + w'xy'z + w'xyz' + wx'yz' + wxy'z$$

Laws of Boolean algebra

These can often help to simplify complicated expressions.

Idempotent Laws

$$(a) \quad a + a = a$$

$$(b) \quad a \cdot a = a$$

Complement Laws

$$(a) \quad a + a' = 1$$

$$(b) \quad a \cdot a' = 0$$

Involution Law

$$(a')' = a$$

Associative Laws

$$(a) \quad (a + b) + c = a + (b + c)$$

$$(b) \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

Commutative Laws

$$(a) \quad a + b = b + a$$

$$(b) \quad a \cdot b = b \cdot a$$

Laws of Boolean algebra

Distributive Laws

$$(a) \quad a + (b \cdot c) = (a + b) \cdot (a + c)$$

$$(b) \quad a \cdot (b + c) = a \cdot b + a \cdot c$$

De Morgan's Laws

$$(a) \quad (a + b)' = a' \cdot b'$$

$$(b) \quad (a \cdot b)' = a' + b'$$

Identity Laws

$$(a) \quad a + 0 = a$$

$$(b) \quad a + 1 = 1$$

$$(c) \quad a \cdot 0 = 0$$

$$(d) \quad a \cdot 1 = a$$

Some laws look like ordinary arithmetic, which makes them easy to remember, while others look quite different from ordinary arithmetic.

Using the laws: a note

The commutativity law says:

(i) in any product, the order of multiplication is unimportant.

e.g. $xy'z = y'xz = zy'x = \dots$ etc.

(ii) in any sum, the order of addition is unimportant.

e.g. $x' + y' + z = y' + x' + z = z + y' + x' = \dots$ etc.

The distributive law says:

$$x(y + z) = xy + xz$$

which can be used to remove brackets

It can also be extended to cases such as:

$$(w + x)(y + z) = wy + wz + xy + xz$$

and

$$x(w + y + z) = xw + xy + xz.$$

Grouping the laws differently

Laws involving 0:

$$xx' = 0$$

$$0x = 0$$

$$0 + x = x$$

Laws involving 1:

$$x + 1 = 1$$

$$1x = x$$

$$x + x' = 1$$

Three more useful laws

$$x'' = x$$

$$x + x = x$$

$$xx = x$$

Grouping the laws differently

Examples

1. $y'(yz + xz') = y'yz + xy'z' = 0z + xy'z' = 0 + xy'z' = xy'z'$

2. $xy + xy + wz = xy + wz$

3. $wx'yz = wx'yz = wx'yz$

Using the laws

De Morgan's laws

$$(x + y)' = x'y' \qquad (xy)' = x' + y'$$

These are very important and are used often. They can also be extended to cases with more variables like:

$$(w + x + y + z)' = w'x'y'z'$$

and

$$(wxyz)' = w' + x' + y' + z'$$

Some examples using de Morgan's laws

Typically we will use de Morgan's laws to remove brackets from certain expressions so as to obtain a sum of products.

e.g. $(pq)' = p' + q'$

$$(x'yz)' = x + y' + z'$$

$$(wxy'z')' = w' + x' + y + z$$

e.g. $(p + q)' = p'q'$

$$(x + y' + z)' = x'yz'$$

$$(w' + x' + y + z')' = wxy'z$$

Some examples using de Morgan's laws

Exercise: Reduce each of the following expressions to a sum of products.

(Note that sometimes it may be necessary to introduce brackets during the simplification.)

1. $(x + y'z)'$

2. $(xy' + yz')'$

3. $(x' + xy'z + yz')'$

4. $(xy' + x'z + yz')'$

5. $(x' + yz)'(wy' + xz)'$

Some examples using de Morgan's laws

Example 1

$$(x + y'z)' = x'(y'z)' = x'(y + z') = x'y + x'z'$$

Some examples using de Morgan's laws

Example 1

$$(x + y'z)' = x'(y'z)' = x'(y + z') = x'y + x'z'$$

Example 2

$$\begin{aligned}(xy' + yz')' &= (xy')'(yz')' = (x' + y)(y' + z) \\ &= x'y' + x'z + yy' + yz \\ &= x'y' + x'z + yz\end{aligned}$$

Some examples using de Morgan's laws

Example 3

$$(x' + xy'z + yz')' = x(xy'z)'(yz')'$$

Some examples using de Morgan's laws

Example 3

$$\begin{aligned}(x' + xy'z + yz')' &= x(xy'z)'(yz')' = x(x' + y + z')(y' + z) \\&= (xx' + xy + xz')(y' + z) \\&= (xy + xz')(y' + z) \\&= xyy' + xyz + xy'z' + xzz' \\&= xyz + xy'z'\end{aligned}$$

Some examples using de Morgan's laws

Example 4

$$\begin{aligned} (xy' + x'z + yz')' &= (xy')'(x'z)'(yz')' = (x' + y)(x + z')(y' + z) \\ &= (xx' + x'z' + xy + yz')(y' + z) \\ &= (x'z' + xy + yz')(y' + z) \\ &= x'y'z' + x'zz' + xyy' + xyz + yy'z' + yzz' \\ &= x'y'z' + xyz \end{aligned}$$

Some examples using de Morgan's laws

Example 5

$$\begin{aligned}(x' + yz)'(wy' + xz')' &= x(yz)'(wy')'(xz')' \\&= x(y' + z')(w' + y)(x' + z) \\&= x(x' + z)(y' + z')(w' + y) \\&= (xx' + xz)(w'y' + yy' + w'z' + yz') \\&= xz(w'y' + w'z' + yz') \\&= w'xy'z + wxzz' + xyzz' \\&= w'xy'z\end{aligned}$$

Expanding a SOP into a complete sum of products

Sometimes it is desirable to expand a sum of products into a **complete sum of products**.

In a complete sum of products each term must be a **fundamental product**; that is, in each term **every variable must be present**, with or without a complement.

If any term of a sum of products has a variable missing, we multiply the term by 1 in the form $(\text{var} + \text{var}')$.

Since the sum $(\text{var} + \text{var}') = 1$, we are multiplying by 1, which does not alter the value of the original expression.

Expanding a SOP into a complete sum of products

Exercise: Write each expression as a complete sum of products

1. $P(x,y,z) = xy'$

2. $M(x,y,z) = z$

3. $L(w,x,y,z) = wxy'z + xyz' + w'yz$

Expanding a SOP into a complete sum of products

Example 1:

$$P(x,y,z) = xy' = xy'(z + z') = xy'z + xy'z'$$

Example 2:

$$\begin{aligned} M(x,y,z) = z &= z(x + x') = xz + x'z = xz(y + y') + x'z(y + y') \\ &= xyz + xy'z + x'yz + x'y'z \end{aligned}$$

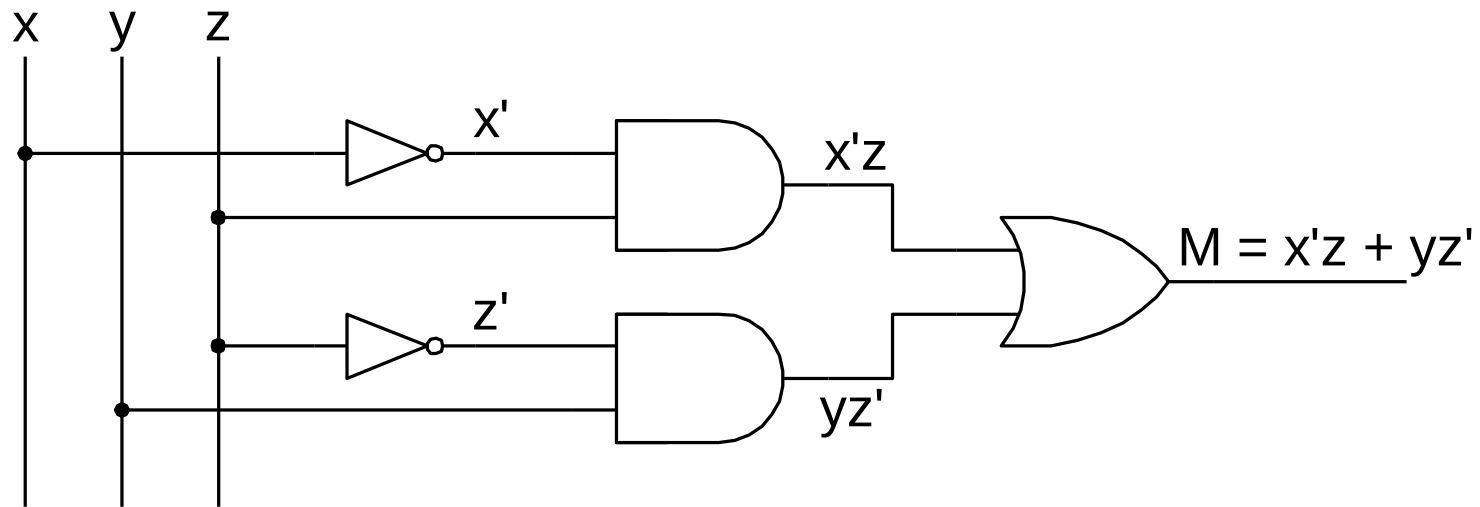
Example 3

$$\begin{aligned} L(w,x,y,z) &= wxy'z + xyz' + w'yz = wxy'z + xyz'(w + w') + w'yz(x + x') \\ &= wxy'z + wxyz' + w'xyz' + w'xyz + w'x'yz \end{aligned}$$

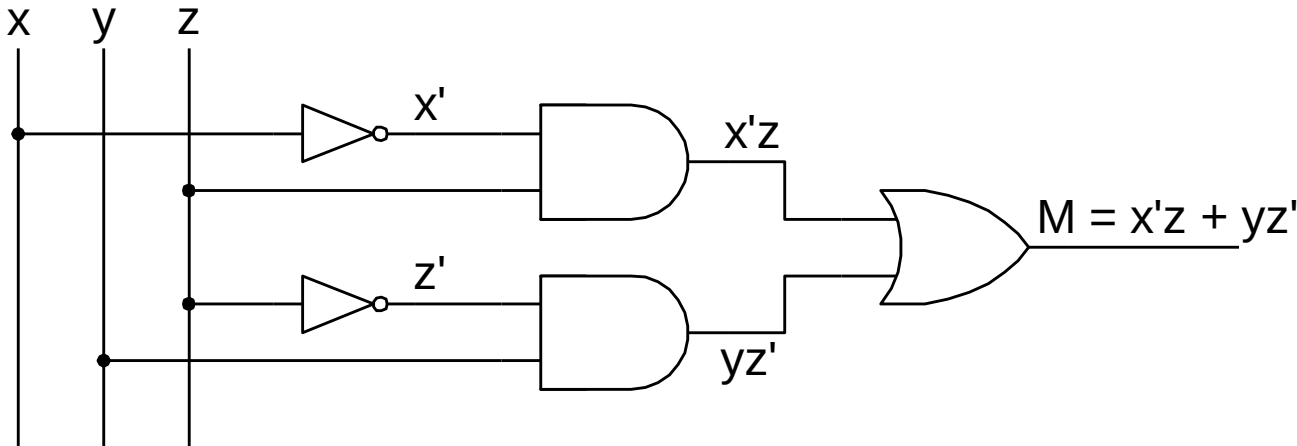
Expanding a SOP into a complete sum of products

Exercise 1

For which combinations of x, y and z will M be equal to 1?



Expanding a SOP into a complete sum of products



Solution:

Expand M into a complete sum of products.

$$M = x'z + yz' = x'yz + x'y'z + xyz' + x'yz'$$

so

$$M = 1 \text{ for } (x,y,z) = (0,1,1), (0,0,1), (1,1,0) \text{ and } (0,1,0)$$

The End